ME 4638: Computational Mechanics Lab

Assignment IV & V

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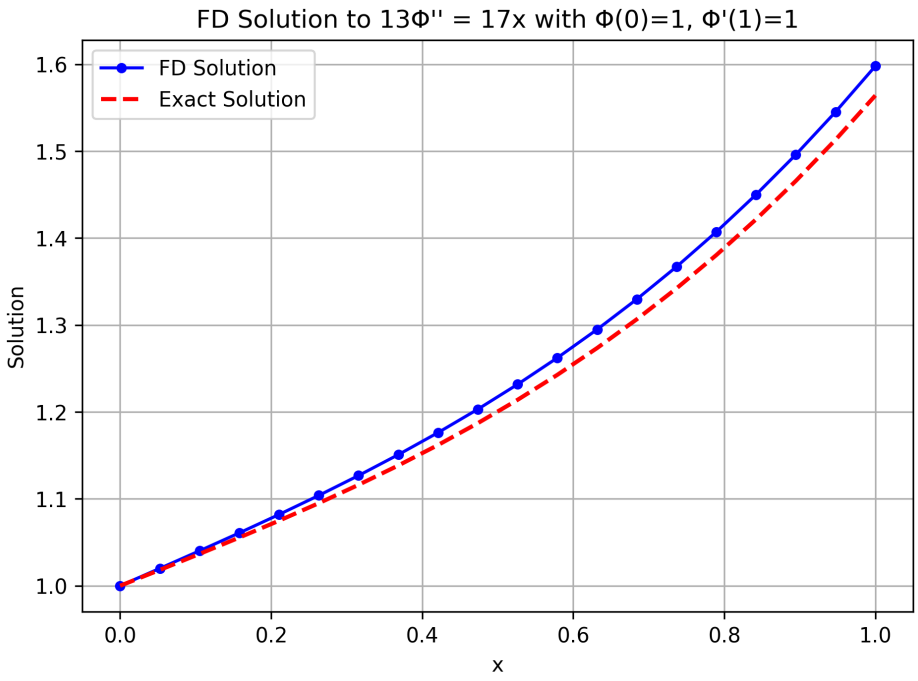
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Date of Submission: October 15, 2026

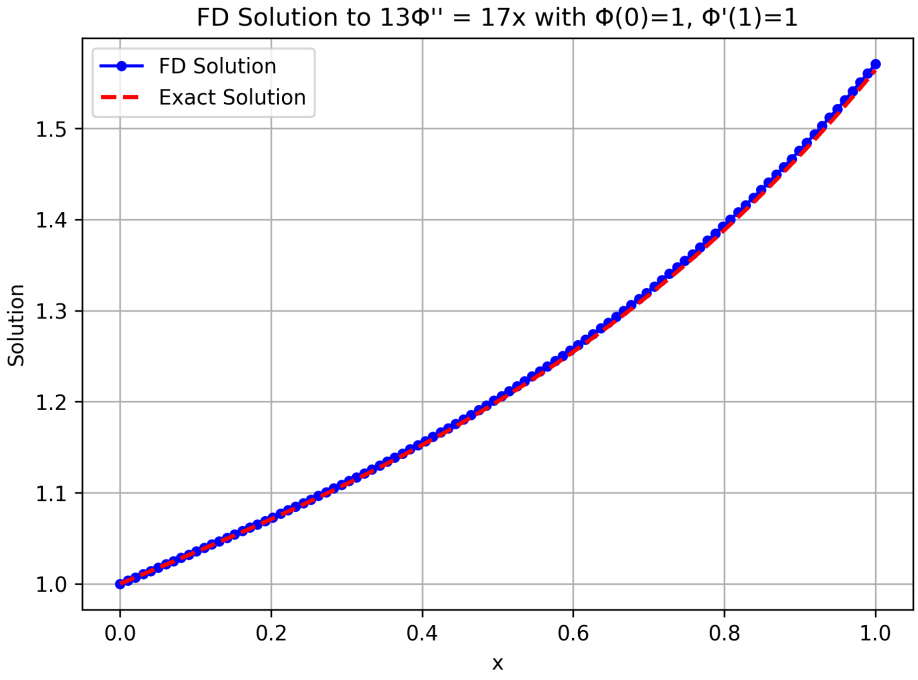
Experiment 4

Problem I

No. of node: 20



No. of node: 100



Problem II

As the number of node increase, the numerical solvers becomes able to capture the tiny or small relationship in the equations like the one in graph or in calculus where the smaller the dx, the refiner or exact the solutions.

Problem III

In neumann boundary condition, we know the exact value and that matches with our results in case of second one, when the number of node was 100. But, for dirichlet boundary condition, as we do not know that exact value, rather we have the differentiation form. This matches when our node was 20, and we observed the differentiation does not become zero.

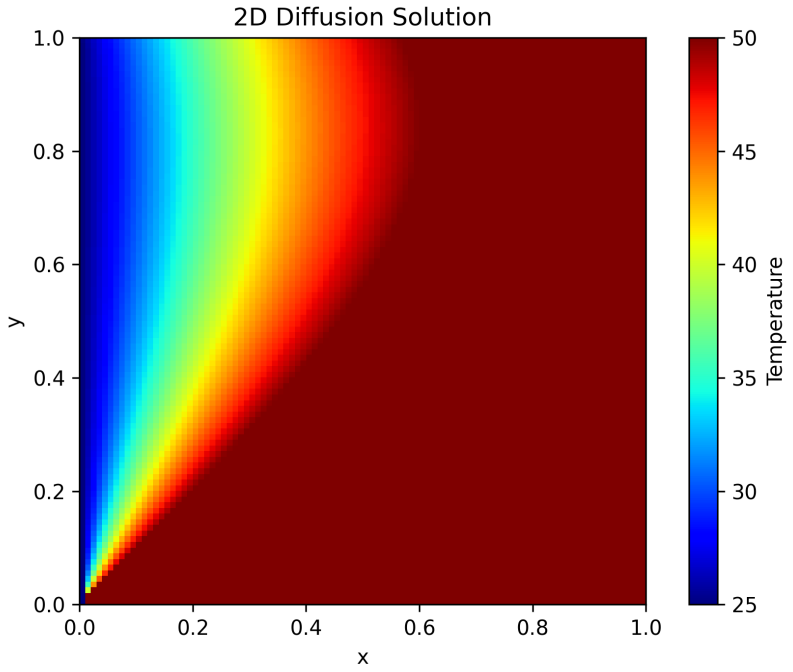
Obviously, the neumann boundary condition provides better accuracy as we know the exact value.

Problem IV

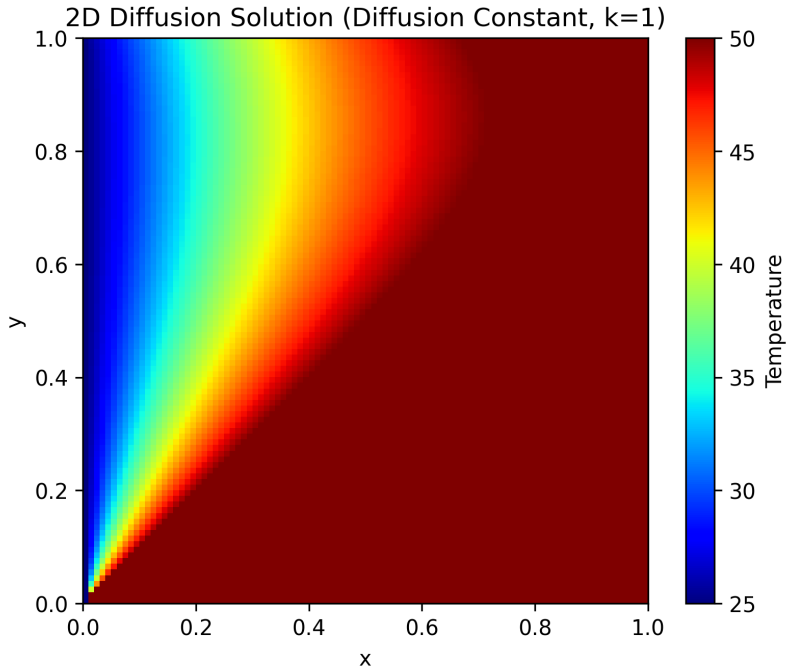
If the boundary conditions were dirichlet boundary condition, then the solution would never converged.

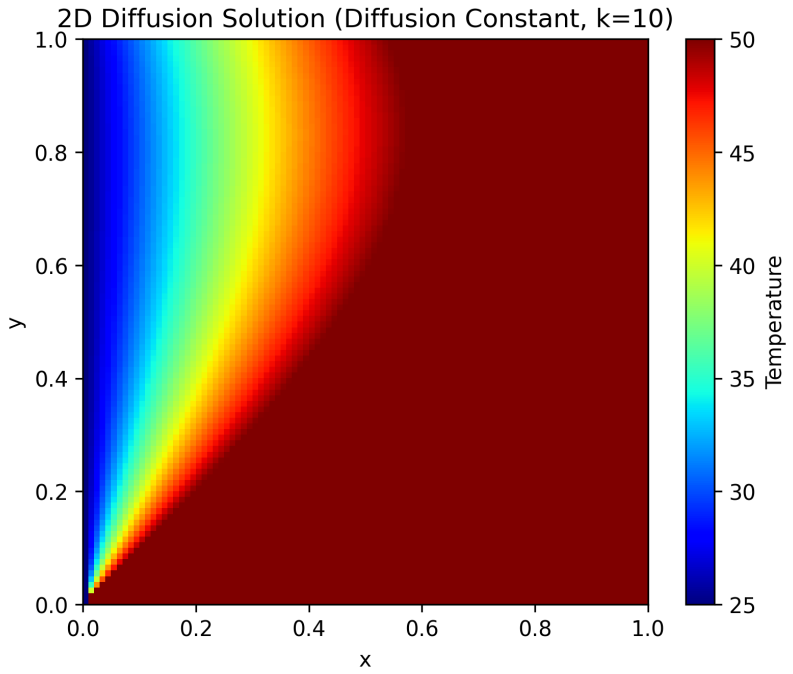
Experiment 5

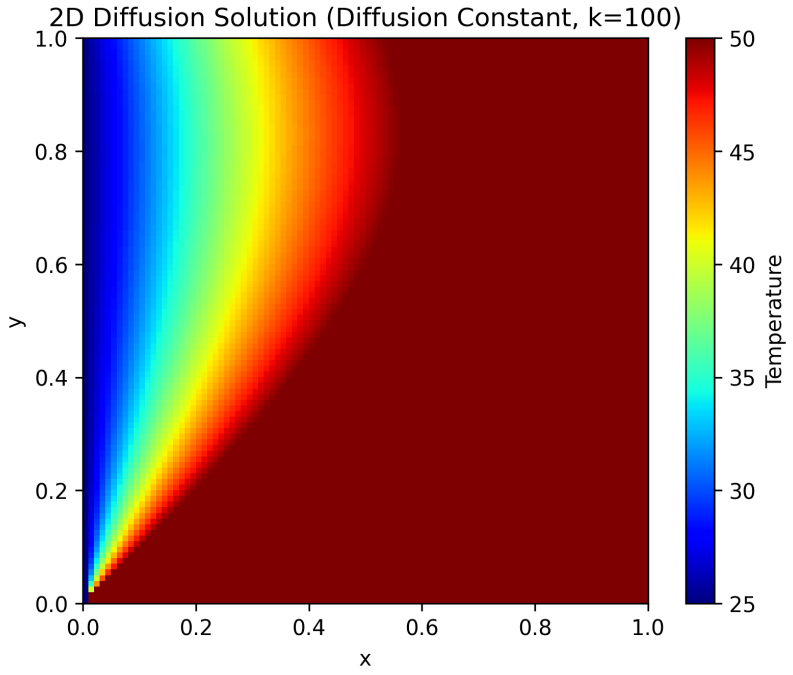
Problem I



Problem II




2D Diffusion Solution (Diffusion Constant, k=10)
1.0
0.8
0.6
0.4
0.2
0.0
y
0.0
0.2
0.4
0.6
0.8
1.0
x
50
45
40
35
30
25
Temperature


2D Diffusion Solution (Diffusion Constant, k=100)
1.0
0.8
0.6
0.4
0.2
0.0
y
0.0
0.2
0.4
0.6
0.8
1.0
x
50
45
40
35
30
25
Temperature